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Authors:

Diogo Baerlocher, University of South Florida
Rodrigo Schneider, Skidmore College

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Diogo Baerlocher
University of South Florida

Rodrigo Schneider
Skidmore College

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Keywords: Racial classification, Political representation, Phenotypic discrimination, Public employment

Baerlocher: Department of Economics, University of South Florida, 4202 E. Fowler Ave. CMC342, Tampa, FL, 33620. (email: baerlocher@usf.edu). *Schneider:* Department of Economics, Skidmore College, 223 Filene Hall, 815 N Broadway, Saratoga Springs, NY, 12866 USA. (email: rschnei2@skidmore.edu). We thank two anonymous reviewers and the editor for helpful comments and suggestions. All remaining errors are our own.

1 Introduction

Racial inequality remains a persistent feature of many democracies, and political representation is often viewed as a critical avenue for addressing it. The race and ethnicity of elected officials are thought to influence political behavior, policy outcomes, and the extent to which historically marginalized groups feel included in the political process. However, racial self-identification is not always straightforward—particularly in societies whose colonial histories led to widespread racial mixing and the formation of social hierarchies that privilege dominant racial groups. In such contexts, racial self-classification may limit the transformative potential of descriptive representation.

In this paper, we examine how racial self-classification among elected city councilors relates to policy outcomes in the Brazilian context. Specifically, we focus on councilors who self-identify as *white* despite possessing phenotypical characteristics that make such classification unlikely. Throughout the paper, we refer to these individuals as *misaligned whites*. To assess phenotypical appearance, we apply a facial classification algorithm to candidate photographs, which assigns probabilities of classification into six ethnoracial categories. Using these outputs, we calculate each candidate's predicted probability of being perceived as *white* and identify those who self-reported as *white* despite having a low predicted probability of being perceived that way. Focusing on the 2020 municipal elections for city councils, we find that municipalities where *misaligned white* candidates were elected by a narrow margin have a lower share of *nonwhite* public employees.

Our empirical strategy relies on a Regression Discontinuity Design, comparing municipalities where *misaligned white* candidates were narrowly elected to the city council with those where *misaligned white* candidates narrowly lost. We show that, near the cutoff, the individual characteristics of the candidates — as well as the characteristics of the municipalities — are similar. The only exception is population size, which is larger in municipalities where *misaligned white* candidates were elected. However, applying RD to multi-member elections raises two identification challenges: a compositional imbalance favoring larger municipalities on the treated side, and an order-statistics concern arising from selecting the marginal candidate based on realized proximity to the cutoff. We address these through propensity score matching, an ex ante candidate selection procedure based on predicted electoral margins, and a range of alternative specifications. Data on the margin of victory, self-reported race, and other socioeconomic characteristics of the candidates come from the Brazilian Superior Electoral Court, while our primary source of data on

public employment is the Annual Report of Social Information (RAIS), maintained by Brazil's Ministry of Economy.

We also present an additional set of analyses to strengthen the validity and interpretation of our findings. We show that the effect of electing a *misaligned white* politician is more pronounced among temporary public workers, over whom elected officials have greater hiring discretion, than among permanent workers who possess tenure and are more difficult to replace. This result reinforces the interpretation that the observed effects stem from political actions undertaken by *misaligned white* politicians, rather than from structural features that our empirical strategy could not fully account for. A decomposition by counterfactual type suggests that the effect is driven by differences among white-identified candidates—specifically, between *misaligned* and *aligned whites*—rather than by the displacement of nonwhite officeholders.

Finally, we examine changes in leadership by comparing the racial composition of the heads of the health and education secretariats before and after the 2020 election. We find suggestive evidence that electing one additional *misaligned white* politician reduces the probability that a municipality experiences an increase in the number of *nonwhite* managers in the health secretariat, but we find no significant effect for the education secretariat. However, the estimated effects on the heads of the secretariats are sensitive to specification choices and should be interpreted with caution.

A key feature of the Brazilian context is the fluidity of racial classification. Since 2014, political candidates have been required to report their race, yet self-identification often diverges from social perception. For example, Janusz (2022) finds that roughly one-quarter of politicians in their sample reported a racial group different from the one perceived by a panel of coders. Further, Janusz (2021) documents that 27% of politicians who ran in both 2014 and 2016 changed their self-reported race, and Romero (2021) reports that over 4,000 candidates reclassified themselves from *white* to *black* in the 2020 elections. These patterns highlight how, in Brazil, race operates as a social construct (Sen and Wasow, 2016).

Brazil's understanding of race is deeply rooted in its unique historical formation. Portuguese colonization, African slavery, and the presence of Indigenous peoples produced a broad range of phenotypical characteristics, which form the main basis for racial classification in Brazil.[1] After the abolition of slavery in the late nineteenth century, the state pursued a "whitening" policy, notably removing the *pardo* category from the 1890 Census and classifying mixed-race individuals as *white* to inflate the proportion of *whites*

[1] Unlike in the U.S., where race is largely determined by ancestry (Skidmore, 2009, p. 5).

(Camargo, 2009). Historically, this contributed to mixed-race Brazilians' tendency to self-identify as *white*.[2] Baerlocher, Caldas and Cavalcanti (2026) find that this tendency is stronger in areas with larger presence of slaves in 1872. From the 1990s onward, Afro-Brazilian organizations advanced self-identification based on ethnoracial identity, and affirmative action policies reinforced this movement. As a result, the 2022 Census, for the first time, reported a majority identifying as *pardo* (Pinhoni and Croquer, 2023).

Given the Brazilian context, one possible way to interpret the results is through councilors' racial group consciousness, defined as the realization of one's position as a disadvantaged racial group and the willingness to use politics to improve this position (Smith, Clemons, Krishnamurthy, Martinez, McLaren and White, 2024, p. 1672).[3] Racial group consciousness thus comprises two components: racial awareness and a willingness to act on behalf of the group. *Misaligned white* self-identification may arise either because individuals assign limited importance to racial distinctions, thereby reproducing historical tendencies toward whitening in Brazil, or because they recognize racial disadvantage but nonetheless identify with a socially dominant racial category.[4] Our empirical design does not distinguish between these mechanisms. However, both imply that the election of individuals whose phenotypical appearance aligns with disadvantaged racial groups need not translate into actions that enhance representation of those groups.

Related Literature Our paper contributes to the literature on the effects of ethnoracial representation on constituents. For example, Beach and Jones (2017) show that greater ethnic diversity in California city councils leads to reduced public spending, likely due to increased disagreement among legislators—consistent with the classic findings of Alesina, Baqir and Easterly (1999). In a related study, Beach, Jones, Twinam and Walsh (2024) find that the election of *nonwhite* city council members led to larger gains in housing values in *nonwhite* neighborhoods compared to *white* ones. Focusing on labor markets in large U.S. cities, Nye, Rainer and Stratmann (2015) show that *black* employment and labor force participation increase following the election of *black* mayors.

[2]This tendency helped prompt the removal of race-related questions from several censuses. As Camargo (2009) notes, the 1920 Census omitted a race question altogether, while the 1940 Census replaced it with a "color" question and excluded *pardo*. Since 1950—except in 1970—the census has included color (and "color and race" since 1991) with *white*, *black*, *yellow*, and *brown* (*pardo*) as main categories.

[3]While prior research has focused on how such consciousness shapes voter behavior — often with reference to the American Civil Rights Movement (*e.g*, Cascio and Washington, 2014) — we shift the lens to politicians themselves.

[4]It is unlikely that candidates gain electoral advantage from how they self-report race. If race matters to voters, it is more likely inferred from candidates' appearance rather than from official classifications, since ballots do not display candidates' self-reported race, and voters typically observe candidates through campaign materials, television advertisements, debates, and in-person campaigning. For this reason, we do not view misaligned racial self-identification as a strategic signal aimed at voters.

In the Brazilian context, de Lucena Coelho, Estevan, Nakaguma and Rabelo (2024) find that the election of *nonwhite* mayors does not significantly affect the employment of *nonwhite* individuals in top government positions, whereas Ikawa, Martins, Sant'Anna and Santarrosa (2024) shows that the election of *black* mayors leads to higher enrollment and graduation rates among *black* students. The study by de Lucena Coelho et al. (2024) is particularly relevant to ours, as it also examines the impact of local elections on public employment and relies on candidate photographs. However, unlike our approach, they use photographs to infer candidates' race in the absence of self-reported data, while we use photographs to identify inconsistencies between self-reported race and phenotypical characteristics. Another key difference is that we focus on the election of city council legislators, whereas both cited studies examine mayoral elections.

We focus on city councilors rather than mayors for two main reasons. First, recent research highlights the significant role of councilors in shaping local governance. They exert direct influence over temporary hiring within their jurisdictions and indirect influence over executive appointments—such as municipal secretariats—by leveraging their control over budget approval and coalition support to secure politically valuable contracts. For instance, Colonnelli, Prem and Teso (2020) shows that discretionary public employment in Brazil serves as a patronage resource. Other studies demonstrate that councils shape fiscal outcomes through their authority over taxation and spending (Schneider and Veras, 2023), and can even affect homicide rates, despite security being a state-level responsibility (Novaes, 2024). Second, our analysis of councilors is motivated by sample size considerations. Because information on the race of public employees is only available for 2023, we must link employment outcomes to the 2020 elections. This makes it difficult to rely solely on mayors, whose numbers are limited once restrictions related to phenotypical appearance are applied. Focusing on councilors therefore provides a much larger and more informative sample.

2 Empirical Strategy

Our objective is to estimate the effects of electing *misaligned white* municipal councilors on the racial composition of municipal hiring. We define such a candidate as one who self-reports as *white* but whose phenotypical appearance suggests otherwise. We describe how we identify them in detail below. Our empirical strategy is a cross-sectional regression discontinuity (RD) design:

$$Y_m = \alpha + \lambda D_m + \beta_1 M_m + \beta_2 (M_m \times D_m) + \epsilon_m, \tag{1}$$

where Y_m denotes the racial composition of municipal hires in municipality m, D_m is a binary indicator equal to one if a *misaligned white* candidate was elected, and $M_m \in [-h, h]$ is the margin of victory centered at zero. The error term is ϵ_m, and the bandwidth h is selected using the MSE-optimal procedure of Calonico, Cattaneo and Titiunik (2014).

Since we study legislative elections featuring multiple winners, the construction of the running variable differs from standard close-race studies and deserves detailed explanation. Boas and Hidalgo (2011); Firpo, Ponczek and Sanfelice (2015); Novaes (2024) also study legislative elections in Brazil. We follow Novaes (2024) closely in the construction of our running variable.

Victory in municipal council elections in Brazil is determined by open-list proportional representation.[5] Under this system, seats are first allocated to parties using the D'Hondt method and then distributed within each party according to candidate vote share. To formalize, consider municipality m with P_m parties indexed by $j \in \mathcal{J}_m = \{1, \ldots, P_m\}$. Let s_{jm} denote the number of seats won by party j, and let N_{jm} denote the number of candidates fielded by that party, indexed by $i \in \mathcal{I}_{jm} = \{1, \ldots, N_{jm}\}$ and ranked by vote share. Denote the vote share of candidate i from party j as $v_{jm}(i)$, with $v_{jm}(1) \geq v_{jm}(2) \geq \ldots \geq v_{jm}(N_{jm})$. Thus, candidate i wins a seat if and only if $i \leq s_{jm}$—that is, if the candidate ranks among the top s_{jm} vote-getters within the party.

Let $\mathcal{W}_{jm} = \{i \in \mathcal{I}_{jm} : \text{candidate } i \text{ is misaligned white}\}$ and $\mathcal{W}^*_{jm} = \{i \in \mathcal{W}_{jm} : i \leq s_{jm}\}$ be the set of *misaligned white* candidates and winning *misaligned white* candidates in each party-municipality pair, respectively. Moreover, let $\mathcal{T}_m = \{j \in \mathcal{J}_m : \mathcal{W}^*_{jm} \neq \emptyset\}$ be the set of parties in m that have at least one *misaligned white* winner. We are now ready to define the (signed) running variable as

$$M_m = \begin{cases} \min\limits_{j \in \mathcal{T}_m} \left[v_{jm}(\max \mathcal{W}^*_{jm}) - v_{jm}(s_{jm} + 1) \right], & \text{if } \mathcal{T}_m \neq \emptyset, \\ \max\limits_{j \in \mathcal{J}_m} \left[v_{jm}(\min \mathcal{W}_{jm}) - v_{jm}(s_{jm}) \right], & \text{if } \mathcal{T}_m = \emptyset, \end{cases} \tag{2}$$

which implies $D_m = \mathbf{1}\{M_m \geq 0\}$.

In words, for municipalities with at least one elected *misaligned white* candidate, the running variable is the smallest vote-share difference—across parties—between the last elected *misaligned white* candidate and the first non-elected candidate within the same party. In municipalities with no elected *misaligned*

[5] Prior to 2020, parties could form coalitions for seat allocation in legislative elections. Starting in 2020, coalitions were prohibited, so the elections we study involve only party-level competition.

white candidates, the running variable is the smallest vote-share difference in absolute value—across parties—between the last elected candidate and the highest-ranked *misaligned white* candidate who narrowly lost.

When both winning and losing *misaligned white* candidates are present in the same municipality, the construction of the running variable is anchored to the winning margin, ensuring that treatment status is determined by the closest electoral victory. As a result, the running variable is positive if and only if a *misaligned white* candidate is elected, and its magnitude reflects the closeness of the relevant electoral contest.

To illustrate, consider a municipality with a single party that wins two seats. Suppose the party fields four candidates with vote shares 30%, 25%, 20%, and 15%, where the third-ranked candidate (20%) is *misaligned white*. Since seats go to the top two vote-getters, the *misaligned white* candidate loses. The running variable equals $0.20 - 0.25 = -0.05$, reflecting that the *misaligned white* lost by 5 percentage points. Had the *misaligned white* been second-ranked (25%) and thus elected, the running variable would equal $0.25 - 0.20 = +0.05$, the margin of victory over the first non-elected candidate.

Data. Our primary source for data on the racial composition of municipal hiring is the Annual Report of Social Information (RAIS), maintained by Brazil's Ministry of Economy. The RAIS contains detailed information on formal-sector employees, including wages, hours worked, occupation, sector of employment, and race. While most information is reported by the employer, sociodemographic fields such as race are provided by the worker at the time of hiring. Further details on RAIS collection procedures are discussed in Cornwell, Rivera and Schmutte (2017). We focus on the 2023 data — the third year of the mandate for councilors elected in 2020 and the first year in which race reporting was mandatory in the dataset. Our sample is restricted to municipalities where both *white* and *nonwhite* employees were hired in the legislative sector.[6]

We define *nonwhites* as individuals classified as *black* (*preto*) or *mixed-race* (*pardo*). The other official classifications—*Indigenous* and *Asian* (*amarelo*)—account for less than 3% of legislative-sector workers and are excluded from the analysis, as societal perceptions toward these groups differ from those toward *pretos* and *pardos*.

[6]Race has been an optional RAIS field since 2004 and was often reported in the private sector. However, until 2023, all race entries for public employees were missing (*não identificado*).

Legislative workers are identified using the legal nature (*natureza jurídica*) code 1066, which encompasses both legislative and administrative staff, including councilor aides, analysts, and operational personnel. Our outcome variable is the share of *nonwhite* legislative workers in each municipality. Because race in RAIS is self-reported, this outcome captures changes in reported racial composition, which may differ from appearance-based racial composition.[7] In 2023, municipalities employed on average 57.7% *white* legislative workers.

To complement the hiring data, we incorporate information from the Brazilian Institute of Geography and Statistics (IBGE) on the race of the heads of municipal secretariats for health and education. These data are drawn from the 2018 and 2021 editions of the Municipality Profiles survey (*Perfil dos Municípios*), which reports demographic and organizational characteristics of Brazilian municipalities. Including these earlier years enables us to assess potential political shifts in leadership before the 2023 hiring outcomes. In this exercise, the outcome variable is binary, indicating whether the race of secretariat heads shifted from *white* to *nonwhite* between 2018 and 2021.

The treatment variable is constructed from candidate-level data provided by the Superior Electoral Court (TSE) for the 2020 municipal elections. This dataset contains sociodemographic information for each candidate, including self-reported race. The TSE also provides official photographs of all municipal council candidates, which we process using a facial classification model — *DeepFace* — to infer each candidate's race.[8] This tool employs deep convolutional neural networks trained on large datasets with over 100,000 facial images to estimate the likelihood that an individual belongs to one of six broad racial groups: *asian*, *indian*, *black*, *white*, *middle eastern*, and *latino/hispanic* (Serengil and Ozpinar, 2024).[9] Instead of assigning a single category, the model produces a probability distribution across all six.

[7] Changes in reported racial composition may reflect either appearance-based representation or reported representation. The former has been linked to role-model effects in prior work (*e.g.*, Ikawa et al., 2024), while the latter may operate through subtler channels related to racial classification and perception. Related work shows that individuals may strategically adjust self-classification in response to perceived incentives or discrimination (*e.g.*, Kudashvili and Lergetporer, 2022). While such behavior is conceptually possible in our setting, our data do not allow us to distinguish this channel from appearance-based changes, and there is no direct evidence that such strategic responses drive our results.

[8] We use the *opencv* face-detection algorithm, which is the default option in *DeepFace*. Our main results are robust to employing *RetinaFace* (Deng, Guo, Zhou, Yu, Kotsia and Zafeiriou, 2019) as an alternative, with the sole exception of the findings concerning the heads of secretariats discussed below.

[9] The effectiveness of the method may vary depending on factors such as image resolution, lighting, and the subject's facial orientation. In addition, the algorithm is based on broad, US-oriented racial classifications that do not fully capture Brazil's unique racial dynamics—most notably, the absence of a clear equivalent to the *pardo* category. Nevertheless, because candidate photographs are officially submitted to the Brazilian Electoral Court and must meet minimum quality standards, we expect the algorithm's outputs to be sufficiently reliable. Moreover, the use of broad racial categories aligns with our objective: we are not attempting to assign a definitive racial identity to each individual, but rather to measure the likelihood they are perceived as *white* given their physical appearance.

We define the probability of being classified as *white* as a weighted combination of the algorithm's race-group probabilities. Denote these probabilities for individual k as $p_{rk} \in [0, 1]$ with $r \in \mathcal{R} := \{w, b, a, l, i, m\}$ representing each possible race. The predicted probability of being classified as *white* is then

$$\mathbb{P}(white)_k = F\left(\sum_{r \in \mathcal{R}} \omega_r p_{rk}\right),$$

where $F : \mathbb{R} \to [0, 1]$ is a function and ω_r are weights assigned to each race's probability. Rather than relying on *ad hoc* assumptions about which algorithm categories map to self-reported whiteness in Brazil, we use a data-driven approach. Specifically, we estimate the weights ω_r from a probit regression where the dependent variable indicates whether the individual self-reported as *white* and the regressors are p_{rk}, excluding the constant to avoid collinearity.

Table 1 shows the probit model estimation representing ω_r. In this case, F is the standard normal cumulative distribution function. As expected, the probability of being *white* is largely determined by larger probabilities of being identified as *white* by the algorithm. *Middle Eastern* is the only other ethnoracial group with positive weight. Large probabilities of being classified as *black* or *indian* strongly reduce the probability of being *white*, which is again expected given the similar skin tone between these two groups.

Table 1: **Probit Regression - Dependent variable** = 1 **if self-reported as *white***

	Coefficient	Standard Error	z	$P > \|z\|$	95% Conf. Interval	
white	1.3504	0.0063	215.85	0.00	1.3381	1.3627
black	-2.2498	0.0198	-113.6	0.00	-2.2886	-2.2110
Asian	-0.4470	0.0108	-41.27	0.00	-0.4683	-0.4258
Middle Eastern	0.3745	0.0095	39.64	0.00	0.3560	0.3930
Indian	-2.2146	0.0285	-77.81	0.00	-2.2704	-2.1588
Latino/Hispanic	-0.4486	0.0068	-66.22	0.00	-0.4619	-0.4353

Probit regression with 529,780 observations. The dependent variable indicates if the individual self-reported as *white*. The regressors are the probabilities of being identified in one of the ethnoracial groups listed in column 1 by the facial classification algorithm. The constant is removed since the probabilities must sum to one.

Figure 1 shows the distribution of the predicted probability of being *white*. Panel A indicates that, as expected, most self-reported *nonwhite* candidates have low predicted probabilities. Specifically, 75% have a predicted probability below 53.6%. Panel B likewise displays the expected pattern for self-reported *white* candidates, with 75% having a predicted probability above 46%. We use this 46% threshold to define

misaligned white candidates — those in the bottom quartile of the predicted probability distribution among self-declared *white* candidates.

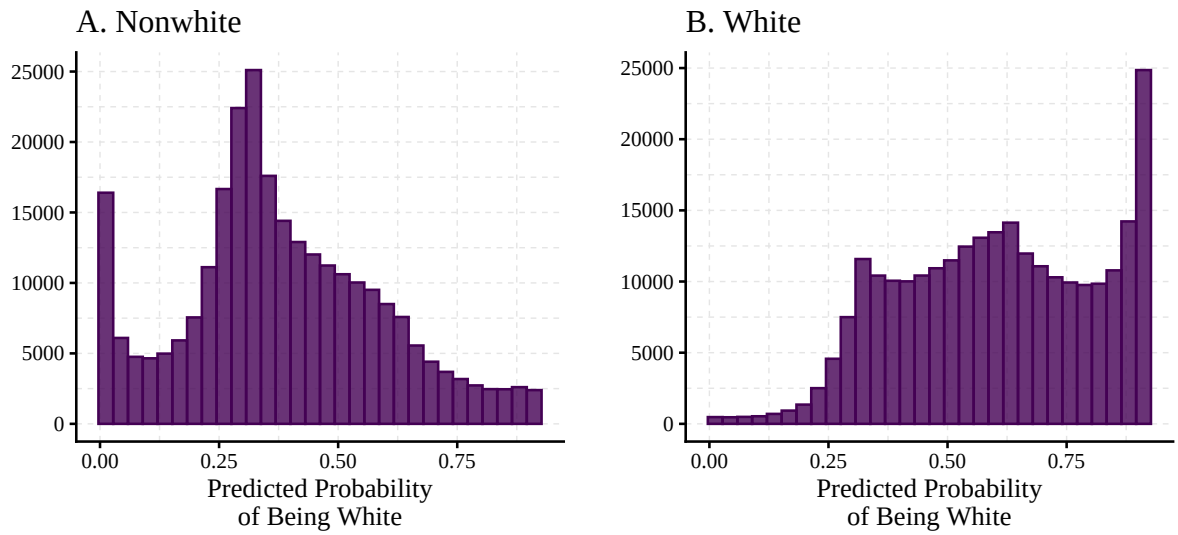

Figure 1: **Predicted Probability of Being *White*.** Distribution of the *predicted probability of being white* for candidates who self-reported as *white* and *nonwhite*. The *predicted probability of being white* are the fitted probabilities from a probit model where the individual's self-reported race is the binary dependent variable and the likelihood that an individual belongs to one of six broad ethnoracial groups as explanatory variables.

Each municipality is associated with a marginal *misaligned white* candidate whose margin of victory equals the municipality's margin of victory according to Equation (2). Table 2 reports summary statistics for municipal council candidates, presented separately for the full sample and the sample of marginal candidates. In the full sample, the majority of candidates are male (about 65%), with an average age of 45. Roughly half are married, and around 23% have higher education. Nearly 47% self-identify as *white*, about 4% are incumbents seeking reelection, and 11% are affiliated with a left-wing party.[10] In the marginal candidate sample, we observe higher proportions of male, married, and higher-educated candidates, as well as a greater share of incumbents. Conversely, the share affiliated with left-wing parties is smaller. By construction, all candidates in this sample self-identify as *white*. Importantly, consistent with our identification strategy, the characteristics in Table 2 show no discontinuous changes at the cutoff, as demonstrated in the next section.

[10]We classify as left-wing the following parties: Workers' Party (PT, the main left-wing party), Socialism and Liberty Party (PSOL), Communist Party of Brazil (PCdoB), Brazilian Communist Party (PCB), Workers' Cause Party (PCO), Unified Socialist Workers' Party (PSTU), Green Party (PV), and Sustainability Network (REDE). The last two parties, PV and REDE, are primarily concerned with environmental causes.

Table 2: **Summary Statistics: Full vs. Marginal Sample**

	Full Sample			Marginal Sample		
Variable	Obs	Mean	Std. Dev.	Obs	Mean	Std. Dev.
Male	487,431	0.653	0.476	3,542	0.872	0.334
Age	487,425	45.280	11.570	3,542	45.511	10.478
Married	487,431	0.503	0.500	3,542	0.621	0.485
Higher Educ.	487,431	0.228	0.419	3,542	0.276	0.447
White	487,431	0.473	0.499	3,542	1.000	0.000
Reelection	487,431	0.043	0.202	3,542	0.161	0.367
Leftwing	487,431	0.115	0.319	3,542	0.077	0.267

Full sample includes all candidates in the sample. Marginal sample includes only the candidates that determine the running variable of each municipality.

3 Interpretation of λ

The parameter of interest in Equation (1) is λ, which under standard RD assumptions measures the marginal effect of electing *misaligned white* councilors. Because we study legislative elections with multiple candidates and seats, the counterfactual to electing a *misaligned white* candidate is electing a non-misaligned candidate whose racial classification varies across municipalities. Empirically, when a *misaligned white* narrowly loses, the elected candidate is an *aligned white*—a self-identified white with high predicted whiteness—in 57% of cases and self-identifies as *nonwhite* in the remaining 43%. Accordingly, our estimates should be interpreted as the effect of adding one *misaligned white* councilor to the overall composition of the legislature, regardless of the specific identity of the candidate who would otherwise have been elected.

Figure 2 clarifies this point. It shows that municipalities in which a *misaligned white* candidate wins by a narrow margin (positive margin of victory) exhibit a substantially higher share of *misaligned white* members in the municipal council than municipalities in which such a candidate narrowly loses (negative margin of victory). Of note, the share of *misaligned white* councilors is calculated from the full composition of the municipal council, not just those within a narrow margin of victory. We observe a discrete increase from approximately 7% to 22% in the share of council seats held by *misaligned white* candidates at the cutoff. This 15-percentage-point discontinuity reflects quasi-random variation in electoral outcomes around the threshold.

Thus, following Firpo et al. (2015), we provide a scaling interpretation by relating the reduced-form effect on public employment to the discontinuity in *misaligned white* representation in municipal councils. We

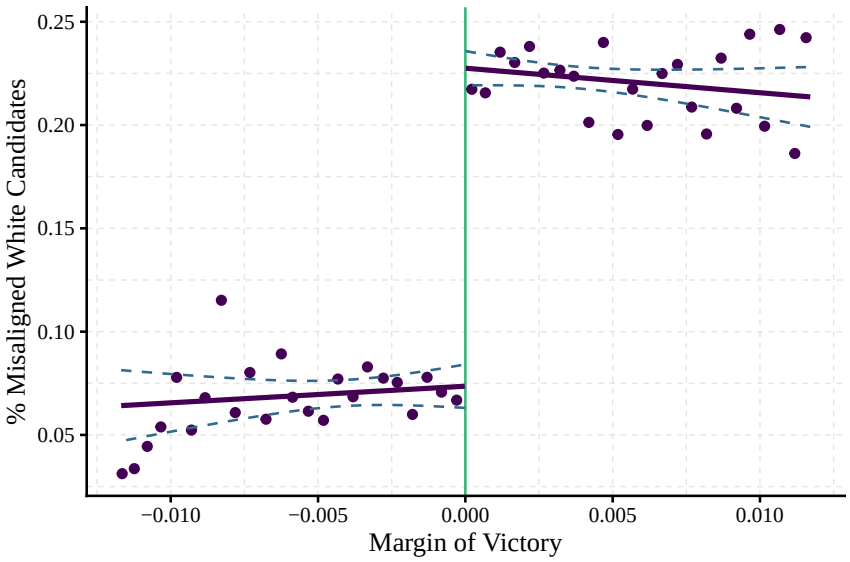

Figure 2: **Discontinuity in Race-Classification Composition.** This figure shows the share of *misaligned white* councilors as a function of the margin of victory. The vertical line marks the cutoff at zero. A solid line is fitted separately on each side of the threshold using MSE-optimal bandwidth with a uniform kernel, and dashed lines show the 95% confidence interval. Scatter points display bin averages. The discontinuity at the cutoff represents the first-stage effect used to compute the Wald estimate.

follow the authors in calling this value the Wald estimate, which measures the effect on the share of *nonwhite* public employees of moving from a hypothetical scenario with no *misaligned white* officeholders to one in which all council seats are held by *misaligned white* candidates. This interpretation extrapolates linearly from the local variation identified at the cutoff and should therefore be understood as illustrative rather than literal. Nonetheless, it highlights the magnitude of the employment effects implied by the observed discontinuities in political representation.

3.1 Causal Interpretation of λ

The construction of our running variable in Equation (2) raises two concerns we must address before interpreting λ as causal. The first concern arises from how we handle municipalities with both winning and losing misaligned white candidates: by anchoring the running variable to the winner's margin, we mechanically assign such municipalities to the right-hand side of the cutoff. Because larger municipalities field more candidates and are thus more likely to have multiple misaligned white candidates, this rule increases the likelihood that larger municipalities appear on the treated side.

This concern finds support in two empirical analyses presented below. First, we implement a manipulation

density test using the procedure of Cattaneo, Jansson and Ma (2020). The test reveals a discontinuity in the density at the threshold. We also present the corresponding density plot in Figure 3A, which includes binned estimates of the running variable's distribution and local polynomial fits with valid confidence bands. Standard RD designs associate such discontinuities with strategic sorting or manipulation around the cutoff, but in our case this is a mechanical feature of the construction of the running variable.[11]

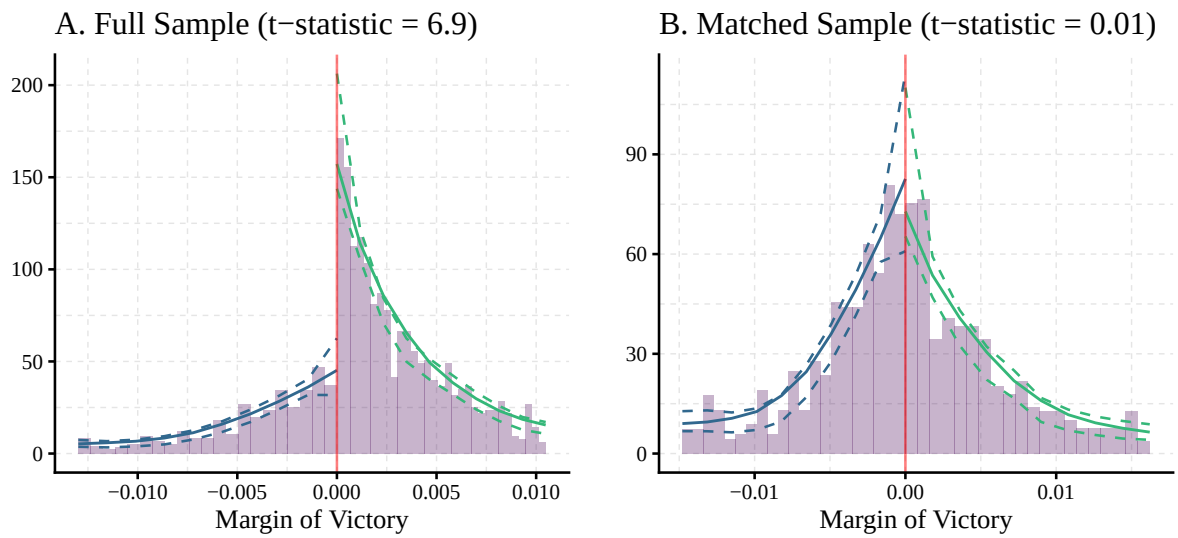

Figure 3: **Margin of Victory Density.** Density of margin of victory around the cutoff zero with bandwidth selected using the uniform kernel function to estimate local polynomial estimators, represented by solid lines. Dashed lines represent 95% confidence intervals computed using jackknife standard errors.

To reinforce that this result is driven by municipality size, Figure 4 shows that the full sample is balanced around the cutoff along many individual and municipal dimensions, except population size. Specifically, in Figure 4A, we apply our regression discontinuity model to test for discontinuities in key characteristics: share of male candidates, average age, share married, share with higher education, share of incumbents seeking reelection, and share affiliated with left-wing parties. For all these variables, we detect no significant jumps at the cutoff, indicating that our sample construction does not systematically alter the composition of candidates. Figure 4B examines potential discontinuities in broader municipal demographic characteristics. First, we test for changes in the share of *nonwhite* residents and find no significant discontinuities, ruling

[11] As an illustration, suppose a case with only one party in each municipality, $P_m = 1\ \forall m$ and denote the running variable for candidate i in municipality m as $M_{im} = v_{1m}(i) - v_{1m}(s_{1m} + 1)$ if i is a winner and $M_{im} = v_{1m}(s_{1m}) - v_{1m}(i)$, otherwise. Let $D_{im} = \mathbf{1}\{M_{im} \geq 0\}$, then $D_m = \mathbf{1}\{\max_{i \in \mathcal{W}^*_{1m}} D_{im} = 1\}$, i.e., treatment is assigned if the municipality has at least one *misaligned white* candidate. Suppose that, close to cutoff, D_{im} is Bernoulli distributed with winning probability equal to ρ. Then, $\mathbb{P}(D_m = 1) = 1 - (1 - \rho)^{|\mathcal{W}^*_{1m}|}$, where $|\mathcal{W}^*_{1m}|$ is the cardinality of $\mathcal{W}^*_{1m}$. Since larger municipalities field more *misaligned whites*, the probability of being treated is larger in large municipalities.

out population composition as a confounding factor. We also assess discontinuities in the logarithms of per capita legislative spending, per capita GDP, and total population. While we find no significant jumps in legislative spending or GDP, we observe a discontinuity in total population: municipalities to the right of the cutoff have populations approximately 40% larger, indicating an imbalance in our sample.

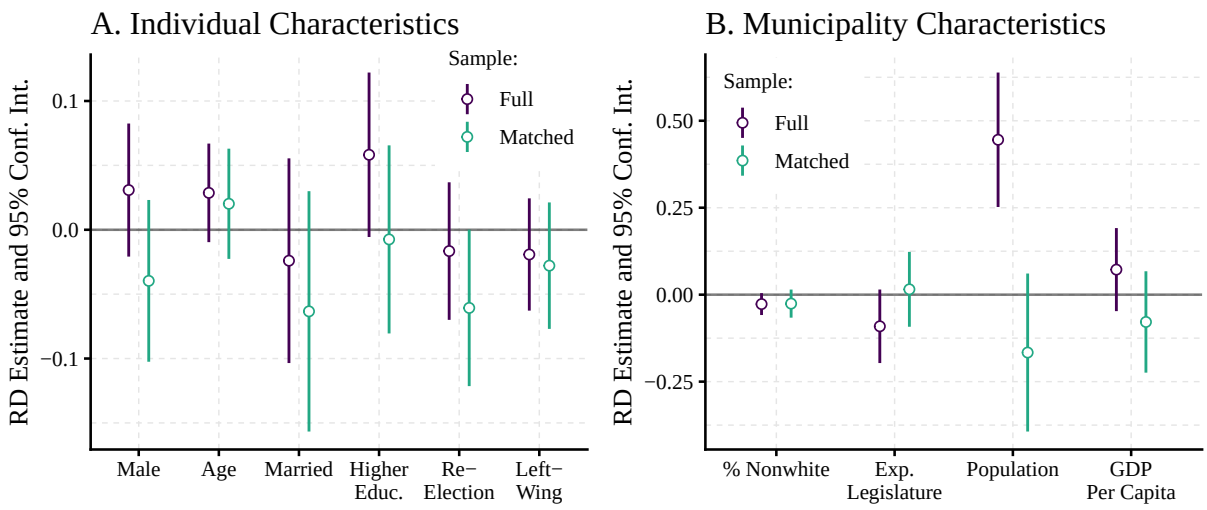

Figure 4: **Balance Tests.** In Panel A, each point represents the estimated effect of electing one additional *misaligned white* councilor on the following characteristics of elected councilors: the share who are male, average age, share who are married, have higher education, are incumbents seeking reelection, and are affiliated with left-wing parties. In Panel B, each point represents the estimated impact of electing one additional *misaligned white* councilor on the following outcomes, respectively: the share of the *nonwhite* population, legislative sector expenditures, total population size, and GDP.

Because municipality size may be correlated with baseline public employment and administrative capacity, this compositional imbalance could in principle affect hiring outcomes. That is, the fact that our running variable is correlated with population size makes the identification assumption $M_m \perp \epsilon_m$ implausible. However, if municipality size is the only possible confounding variable, a version of the identification assumption conditional on population is more plausible. We show that results are similar when controlling for population size, suggesting that this variable has no significant correlation with other variables that determine the racial composition of public employment.

The second concern relates to selection into close contests. Because M_m selects the closest *misaligned white* candidate to the cutoff as the marginal candidate, municipality characteristics that make such candidates systematically more likely to be marginal could bias our estimates, even near the cutoff. This concern is also discussed in Novaes (2024). However, proportional representation systems differ from single-member

plurality races in ways that make such selection less likely. In these elections, closeness depends not only on individual vote shares but also on party-level seat allocations, which are often driven by a small number of highly popular candidates who are far from the cutoff (Boas and Hidalgo, 2011). Moreover, the complexity of seat-allocation rules (e.g., the D'Hondt method), the absence of reliable pre-election polling, and the large number of newcomer candidates make it unlikely that voters can predict or strategically influence which candidates are marginal (Novaes, 2024).

While these institutional features mitigate concerns about systematic selection into close contests, we implement several robustness checks to further guard against selection bias. In our main alternative approach, we match candidates at the left- and right-hand sides of the discontinuity according to a propensity score computed using individual and municipal predetermined characteristics. The density plot for the matched sample is presented in Figure 3B and, together with a manipulation test statistic equal to 0.75, provides evidence of no manipulation around the cutoff. Another important advantage of the matched sample is that it is balanced by construction, as displayed in Figure 4, removing the previous concern about the striking difference in population size. Finally, the matched sample also reduces concerns about selection bias driven by *misaligned white* candidates being more likely to be marginal candidates because we are now comparing a similar set of municipalities.

Section 5 shows that our results are robust to the construction of an alternative running variable based solely on predetermined characteristics and to the exclusion of a "donut" region around the cutoff that eliminates the discontinuity in the running variable's density. Once this region is removed, the test statistic equals 0.9, and we fail to reject the null hypothesis of no manipulation. The results remain robust to this modification, providing further reassurance that our findings are not driven by endogenous selection of marginal candidates.

4 Results

Figure 5 presents our main findings. Panel A shows a clear discontinuity in the share of *nonwhite* employees hired in the legislative sector: municipalities where a *misaligned white* candidate narrowly wins exhibit an 8.4 percentage point lower share than municipalities where such a candidate narrowly loses. Panel B replicates the analysis for the private sector, where no discontinuity is found. This public-private contrast suggests the effect operates through the discretionary hiring authority of elected officials rather than broader

labor market conditions.[12]

The magnitude of this effect is substantial: an 8.4 percentage point decline corresponds to roughly a 20% reduction relative to the average share of *nonwhite* hires. For comparison, existing evidence shows that adding one municipal councilor increases social program expenditures by approximately 16% (Schneider and Veras, 2023)—a comparable magnitude. Results are robust to using the matched sample (Panels C and D), where the point estimate is slightly larger at 9.5 percentage points.

Using the Wald estimator discussed above, we can scale this effect to measure the hypothetical impact of moving from zero to full *misaligned white* representation. The Wald estimate equals $0.084/0.152 = 0.55$. This implies that, under a linear extrapolation, moving from a council with no *misaligned white* members to one composed entirely of such candidates would reduce the share of *nonwhite* public employees by 55 percentage points. This scenario is purely hypothetical and lies far outside the observed variation—in our data, the discontinuity shifts misaligned white representation from roughly 7% to 22%. The Wald estimate thus provides an alternative way to interpret our results by expressing the effect per unit change in council composition.

As noted in Section 3, the counterfactual to electing a *misaligned white* is a mixed set of candidates. To further characterize this counterfactual, we examine what type of candidate is elected when a *misaligned white* narrowly loses. We split the sample based on whether the marginal replacement candidate self-identifies as *white* or *nonwhite*. When the replacement is an *aligned white*, the estimated effect is -0.084 (s.e. = 0.04)—identical to our main result. When the replacement self-identifies as *nonwhite*, the estimate is smaller and statistically insignificant at -0.045 (s.e. = 0.047). Although we cannot reject equality of the two coefficients (p-value = 0.53), this pattern suggests the effect is driven by differences *among white-identified candidates*—specifically, between *misaligned* and *aligned whites*—rather than by the displacement of nonwhite candidates.

We next examine whether the effect varies by job type. Permanent workers in Brazil's public sector obtain tenure after passing competitive exams, making their hiring less susceptible to political influence. Temporary workers, by contrast, can be hired at the discretion of elected officials. Figure 6 shows that the negative effect is more pronounced among temporary employees, consistent with the discretionary hiring channel. Point estimates suggest larger reductions for managerial temporary workers (approximately 25%)

[12]This result reduces the likelihood that the estimates reflect worker-side misalignment driven by shared local environments, as such a mechanism would be expected to affect private-sector and permanent public workers as well.

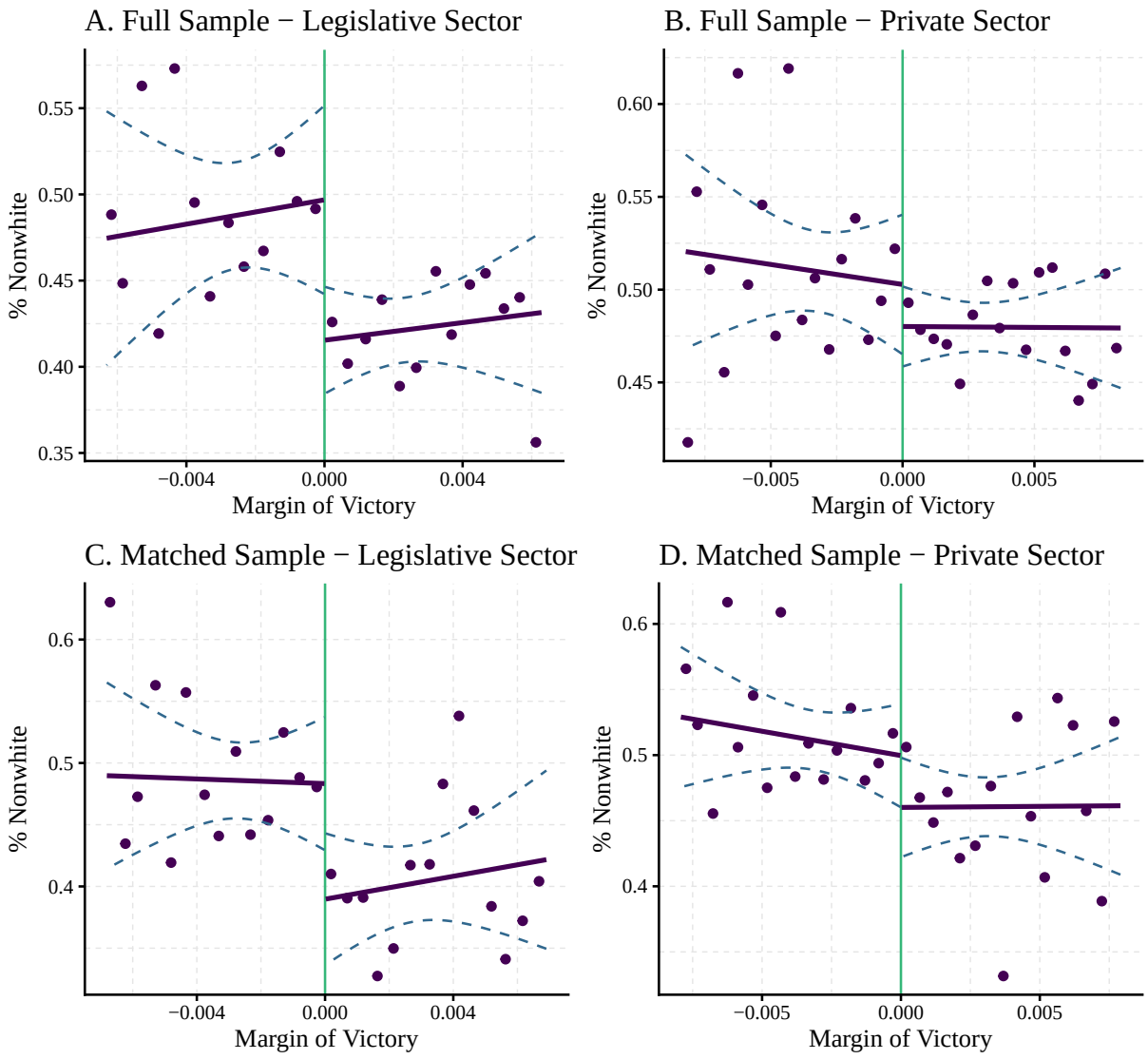

Figure 5: **Margin of Victory Discontinuity.** This figure presents the outcome as a function of the margin of victory, using the MSE-optimal bandwidth approach with a uniform kernel (Calonico et al., 2014). A vertical line marks the cutoff at zero. A solid line is fitted separately on each side of the threshold, and the dashed lines show the 95% confidence interval. Scatter plots display averages within 0.05 percentage point intervals. Panels A and B present results for percentage of *nonwhite* workers hired in the public (legislative) and private sector, respectively. Panels C and D replicate the analysis using the matched sample for the public and private sectors, respectively.

than non-managerial ones (about 20%), though formal difference-in-coefficients tests do not reject equality (temporary vs. permanent: p-value = 0.22; managerial vs. non-managerial: p-value = 0.76). Results are similar in the matched sample, except that the effect on managerial positions is attenuated.

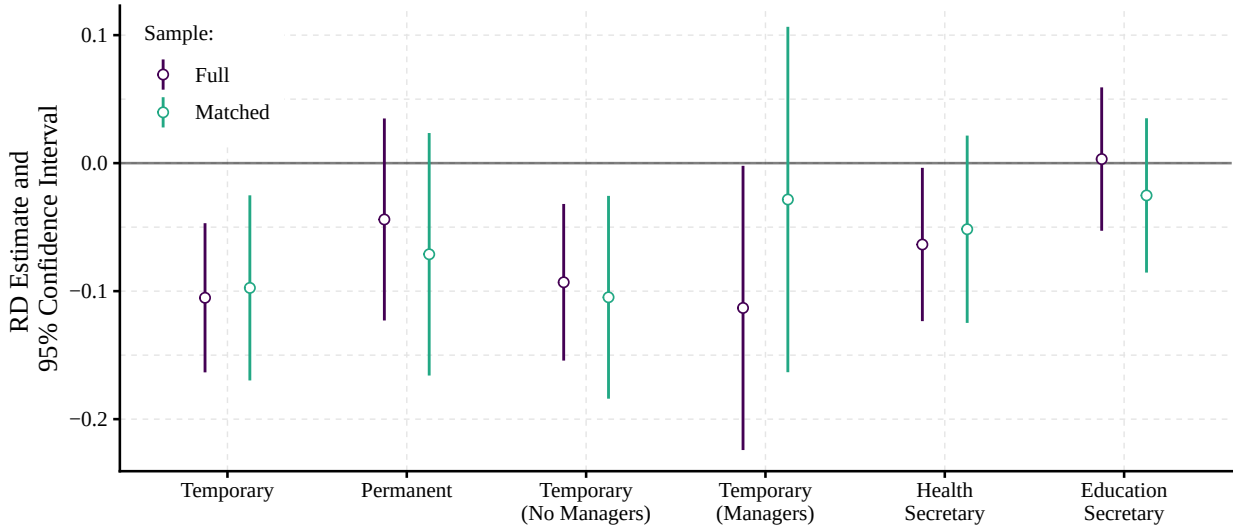

Figure 6: **RD Estimates.** Each point shows the estimated effect of electing an extra *misaligned white* councilor on the percentage of *nonwhite* workers hired in the legislative sector, distinguishing between temporary, permanent, managers and heads of secretariats. In the latter case, the effect is on the likelihood that a municipality experienced an increase in the number of *nonwhite* managers in the health or education secretariats between 2018 and 2021.

Finally, we examine whether effects extend to high-level bureaucratic appointments using IBGE data on municipal secretariat heads. Figure 6 shows that electing a *misaligned white* candidate reduces by roughly 20% the likelihood that the head of the health secretariat shifted from *white* to *nonwhite* between 2018 and 2021. No comparable effect is found for education secretariats. However, these results are sensitive to the choice of face-detection algorithm and less precise in the matched sample, so they should be interpreted with caution.[13]

Taken together, these results indicate that the election of *misaligned white* candidates leads to a substantial reduction in *nonwhite* hiring in the legislative sector. The effect is concentrated in temporary positions where hiring discretion is greatest, and appears to reflect differences in behavior among white-identified candidates rather than the displacement of nonwhite officeholders. In the next section, we assess the robustness of these

[13]When *RetinaFace* is used instead of *opencv*, the health secretariat effect becomes insignificant while the education effect becomes significant. Cross-sectional results using only 2021 data are consistent with our main findings: the coefficient for health secretariats is -0.064 (s.e. = 0.037), and for education secretariats is -0.024 (s.e. = 0.040).

findings to alternative specifications and address the identification concerns discussed in Section 3.

5 Robustness and Extensions

We now assess the robustness of our main findings to alternative specifications, sample definitions, and identification strategies. These checks address the concerns discussed in Section 3—namely, the population size imbalance and potential selection into close contests.

Model Specifications We estimate alternative model specifications to test the robustness of our results and to account for the discontinuity in population size, which could potentially bias our estimates. The results are presented in Figure 7A. All tested models — including those with a second-order polynomial and a triangular kernel — yield similar results. Moreover, adding controls for population size, share of the *nonwhite* population, GDP per capita, and legislative spending increases the magnitude of the estimated effects, indicating that these factors are not driving our findings.

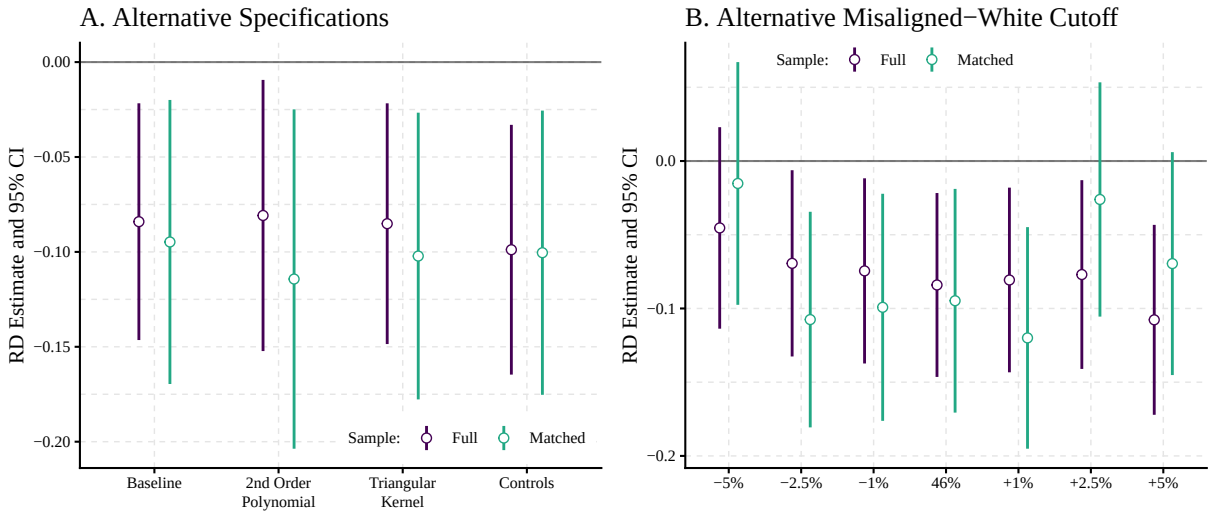

Figure 7: **Alternative Specifications.** Each point represents the estimated impact of electing one additional *misaligned white* councilor on the share of *nonwhite* employees in the legislative sector. We report results from four different model specifications: (i) a baseline model, (ii) a model using a second-order polynomial of the running variable, (iii) a model with a triangular kernel, and (iv) a model that adds controls for the share of the *nonwhite* population, total population size, per capita legislative expenditures, and per capita GDP.

Definition of Misaligned White All estimations so far use our baseline definition of *misaligned white* candidates as those with a predicted probability of being *white* below the first quartile of the distribution (46%). We assess robustness by re-estimating the main specifications using alternative thresholds obtained by adding and subtracting 1, 2.5, and 5 percentage points from the baseline cutoff. The results are in Figure 7B. Expanding the cutoff yields more inclusive samples and estimates similar to the baseline. Tightening the cutoff produces more restrictive samples and generally similar results, though estimates become smaller and less precise in a few cases: the full sample at the 41% threshold, and the matched sample at both the 41% and 48.5% thresholds. These attenuations are consistent with reduced sample size and statistical power rather than a change in the underlying effect.

Donut RDD We re-estimate the manipulation test after excluding observations within a "donut" region of ± 0.005 around the cutoff. Once this region is removed, the test statistic equals 0.9, and we fail to reject the null hypothesis of no manipulation. We thus conclude that the initial jump is localized and not due to strategic manipulation. Our main results remain robust when excluding this donut region. In fact, the estimated effect increases in magnitude and remains statistically significant at the 95% confidence level (coefficient of -16 percentage points, standard error of 7 percentage points), further supporting the credibility of our research design.

Sample We re-estimate all specifications including municipalities that hire exclusively *white* or exclusively *nonwhite* employees. While these cases are highly regionally concentrated and generate corner outcomes in the dependent variable, the estimated effects remain negative and similar in magnitude. Across specifications, coefficients range from -0.060 (s.e. = 0.029) to -0.078 (s.e. = 0.031), compared to -0.084 in the baseline sample. The modest attenuation is consistent with reduced variation and increased heterogeneity rather than a reversal of the effect.

Running Variable A potential concern in open-list proportional representation elections is that selecting candidates based on realized proximity to the electoral cutoff may induce order-statistics bias, as municipalities with larger or more competitive candidate pools could mechanically generate more observations near the threshold. As discussed in Section 3, this concern is attenuated in our setting by the mechanics of Brazil's D'Hondt proportional representation system, where competition for marginal seats occurs among a narrow subset of candidates with tightly clustered vote shares (Boas and Hidalgo, 2011).

Nonetheless, to further address potential order-statistics concerns, we adopt an *ex ante* selection strategy. We first predict candidates' electoral margins relative to the party-level cutoff using only predetermined individual characteristics together with party fixed effects and municipality fixed effects, and define a municipality-specific predicted marginal position. We then select exactly one *misaligned white* candidate per municipality based on proximity to this predicted cutoff, prior to observing realized vote shares. The realized margin of victory is subsequently used solely as the running variable in the RD design. This procedure breaks the mechanical link between candidate pool size and realized proximity to the threshold, ensuring that sample selection is not driven by *ex post* electoral outcomes. Across specifications—including higher-order polynomials, alternative kernel choices, and additional controls—estimated effects remain stable in magnitude and statistically significant at conventional levels; the baseline coefficient is -0.052 (s.e. $= 0.025$).

Labor Market Variables We also examine whether the wage or hours-worked gap between *nonwhite* and *white* employees changes following the election of a *misaligned white* candidate, finding no evidence of such effects in either the legislative or private sectors. These null results suggest that the impact operates on the extensive margin—affecting who is hired—rather than on the intensive margin of job quality or compensation. As an additional mechanism test, we repeat the analysis using candidates who self-identified as *nonwhite* but were classified by the facial classification model as having a high probability of being *white*. In this case, we find no evidence of discontinuities in public hiring by race, corroborating the interpretation that the observed effects are specifically driven by *misaligned whiteness*.

6 Conclusion

This paper provides evidence that the election of *misaligned white* candidates—those who self-identify as *white* but are unlikely to be classified as such by facial classification algorithms—has measurable consequences for racial representation in Brazil's public sector. Using a regression discontinuity design based on narrow electoral victories for municipal councils, we find that the election of such candidates leads to a substantial decline in the hiring of *nonwhite* employees in the municipal legislative sector, amounting to roughly a 20% reduction relative to the average share. The effect appears in the public sector but not in the private sector, and is more pronounced among temporary workers, over whom elected officials have greater

hiring discretion. A decomposition by counterfactual type suggests the decline is driven by differences among white-identified candidates—specifically, between *misaligned* and *aligned whites*—rather than by the displacement of nonwhite officeholders. We also find suggestive evidence that the effect extends to leadership appointments in municipal health secretariats, though these results are sensitive to specification choices.

Applying regression discontinuity methods to multi-member elections raises identification challenges that we address through multiple strategies. The construction of the running variable in proportional representation systems can induce compositional imbalances and order-statistics bias, which we tackle using propensity score matching, an ex ante candidate selection procedure based on predicted electoral margins, and the exclusion of observations near the cutoff. Our main findings are robust across all approaches. The absence of significant effects on wages or hours worked further supports the conclusion that the impact operates on the extensive margin—determining who is hired—rather than on job quality or compensation. Overall, the results highlight how racial self-classification, particularly when misaligned with phenotypical appearance, can shape public employment outcomes through channels of political discretion.

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